longer the duration of the loan because the longer time to repay exposes the lender to greater risk that the borrower may default.

Reversing the concept leads to the conclusion that shorter-term loans should be less risky and therefore require less compensation for the lender. A flash loan is an instantaneous loan paid back within the same transaction. A flash loan is similar to an overnight loan in traditional finance but with a crucial difference: repayment is required within the transaction and enforced by the smart contract.

A thorough understanding of an Ethereum transaction is important for understanding how flash loans work. One clause in the transaction is vital: if the loan is not repaid with required interest by the end of the transaction, the whole process reverts to the state before any money ever left the lender's account. In other words, either the user successfully employs the loan for the desired use case and completely repays it in the transaction, or the transaction fails and everything resets as if the user had not borrowed any money.

Flash loans essentially have zero counterparty risk or duration risk. However, there is always smart contract risk (e.g., a flaw in the contract design; see Chapter 7). Flash loans allow a user to take advantage of arbitrage opportunities or to refinance loans without pledging collateral. This capability allows anyone in the world to have access to opportunities that typically require very large amounts of capital investment. In time, we will see similar innovations that could not exist in the world of traditional finance.